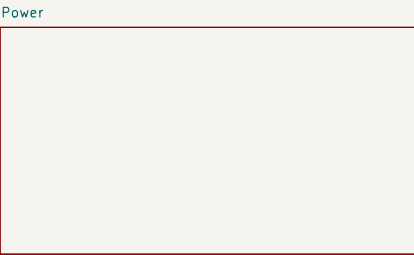
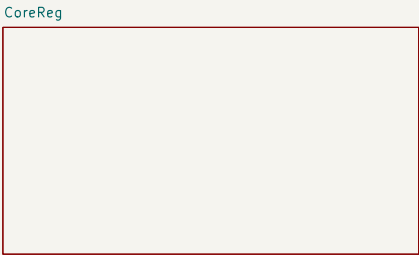


GENERATED FROM test-board.zen -- DO NOT EDIT
Regenerate: python .claude/skills/zen-to-eeschema/scripts/zen_to_sch.py
Nets connect across sheets by GLOBAL LABEL (no sheet pins).

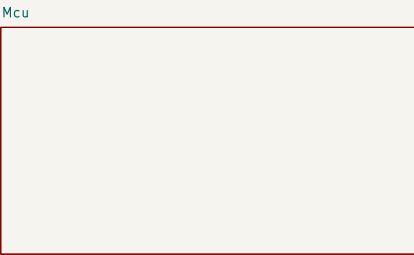
BOARD
test-board
Small RP2350A test board (39 components, 2L, 51x21mm) used as a KiCadRoutingTools test bed.
2 copper layers, 1.00 mm finished
42 components across 11 module sheets, 60 nets, 6 net classes
source 1775e8f34a95-dirty



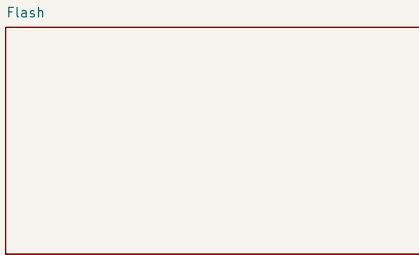
Bestand: Power.kicad_sch



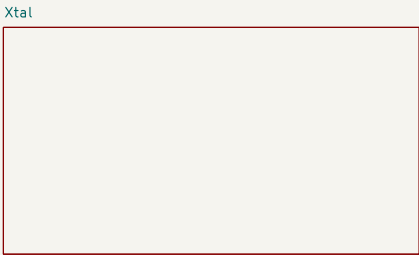
Bestand: CoreReg.kicad_sch



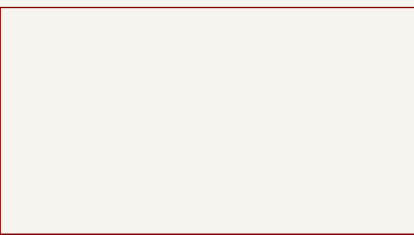
Bestand: Mcu.kicad_sch



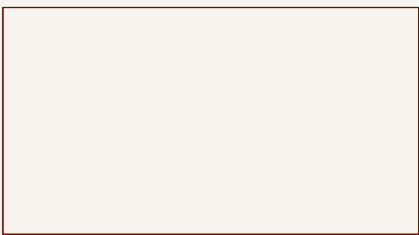
Bestand: Flash.kicad_sch



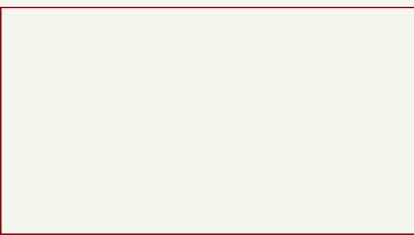
Bestand: Xtal.kicad_sch



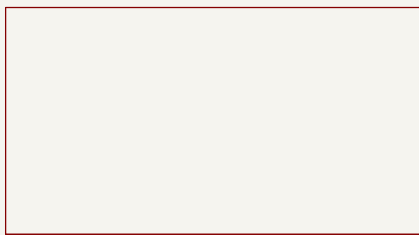
Bestand: Usb.kicad_sch



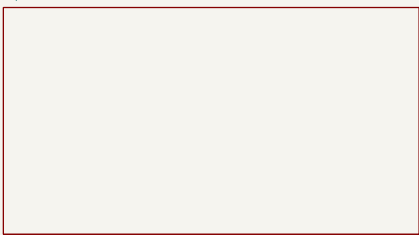
Bestand: Swd.kicad_sch



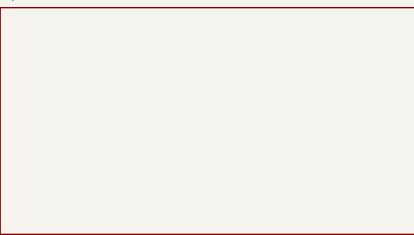
Bestand: Buttons.kicad_sch



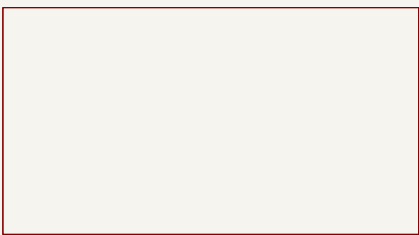
Bestand: Led.kicad_sch



Bestand: GpioA.kicad_sch



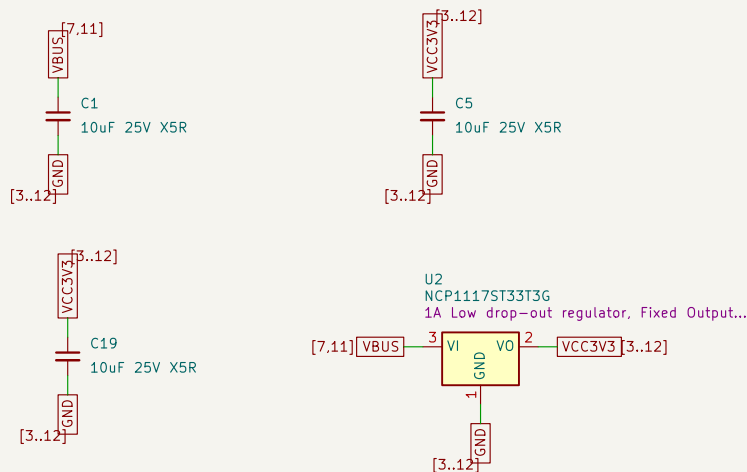
Bestand: GpioC.kicad_sch



Bestand: PartsList.kicad_sch

12 module sheet(s); nets connect by GLOBAL LABEL
source 1775e8f34a95-dirty
GENERATED from test-board.zen by zen-to-eeschema -- DO NOT EDIT
Small RP2350A test board (39 components, 2L, 51x21mm) used as a...
Sheet: /
File: test-board.kicad_sch

Title: test-board -- root		
Size: A2	Date: 2026-08-03T17:42:25+02:00	Rev: 1775e8f34a
KiCad E.D.A. 10.0.0	Id: 1/13	



NET CLASSES ON THIS SHEET

GND -- GND_REF
GND is a B.Cu pour, not a routed trace (HW-TB-PCB24), so track width governs almost nothing. The
class exists to pin GND's stitching vias to 0.6/0.3 and to give the pour an explicit identity.
VBUS, VCC3V3 -- PWR
track width 0.4mm for loop inductance, not current capacity (HW-TB-PCB25). Broader than the
design brief's section 3, which names only VCC3V3 and VBUS.

DESIGN NOTES

HW-TB-COMP02a: the LDO's continuous rating shall exceed the worst-case 3V3 load (41.6mA, hw-spec mid section 4 power budget) by a factor of ≥ 10 .

800mA, not the 1A this check carried until issue #6. The datasheet headline is "output current in excess of 1.0 A", but every electrical characteristic – output voltage, line and load regulation, dropout – is specified to 800mA, and its 1000mA figure is the MINIMUM CURRENT LIMIT (typ 1500, max 2200), i.e. where the part folds back. A fold-back threshold is not a continuous rating. The requirement still passes, at 19.2x against a required 10x.

check: $\text{LDO_I_MAX} \geq \text{RAILL_WORST} * 10$
where $\text{LDO_I_MAX}=800\text{mA}$, $\text{RAILL_WORST}=41.6\text{mA}$

HW-TB-COMP02b: input headroom against the USB VBUS minimum, not its 5V nominal. Datasheet: the 3.3V fixed part is guaranteed over $V_{in} = 4.75\text{V}$ to 10V, dropout 0.95V typ / 1.10V max at $I_{out} = 100\text{mA}$ (the lowest current published; this board draws ~52mA). USB 2.0 section 7.2.2: a high-power port supplies $\geq 4.75\text{V}$ at a full unit load.

CORRECTION: dropout is specified "Measured at $V_{out} - 100\text{ mV}$ ", so the guarantee is $V_{in} \geq 3.20 + 1.10 = 4.30\text{V}$ for $V_{out} \geq 3.20\text{V} - \text{NOT } V_{in} - 3.3\text{V} \geq 1.10\text{V}$. The 4.40V case (a long cable or bus-powered hub) therefore has +100mV of margin, not the "exactly zero" (this comment used to claim, and the binding limit is not the LDO at all: RP2350 Table 1441 puts a 3.135V floor on V_{REG_AVDD} and USB_OTP_VDD , which 3.20V clears by 65mV – the tightest margin on the board. Consequence: a series Schottky would break only the already-marginal 4.40V case, so the dropout budget is NOT a second justification for HW-TB-CONN05's zero VBUS protection; D10's cost/single-source/host-fusing argument stands on its own. See HW-TB-COMP02b.

The check below is deliberately kept in the conservative $V_{in} - V_{out} \geq \text{dropout}$ form: it is a stricter bar than the datasheet's, and passing it means the 4.75V case is clear by inspection.

check: $\text{VBUS_MIN} - \text{VOUT} \geq \text{DROPOUT_MAX}$
where $\text{DROPOUT_MAX}=1.10\text{V}$, $\text{VBUS_MIN}=4.75\text{V}$, $\text{VOUT}=3.3\text{V}$

HW-TB-COMP02c – headroom is budgeted against VBUS MINIMUM, not 5V nominal. At USB 2.0 7.2.2's 4.75V a high-power port leaves 0.35V over the 1.10V max dropout; at the 4.40V floor a bus-powered function must still work at, the headroom is EXACTLY ZERO. Accepted. This is why HW-TB-CONN05 fits zero VBUS protection – a series Schottky would spend the entire margin.

HW-TB-CONN06 – C1's 10uF is the whole USB inrush budget (USB 2.0 7.2.4.1 caps a bus-powered function at 10uF || 440mF). Nothing further goes on VBUS without re-deriving this or adding a soft-start. The other ~40uF on this board sits behind U2's pass element and is charged through its current limit, not through the cable, so it does not count.

HW-TB-COMP02a / D16 – the NCP1117's '4.7uF, ESR 33mohm–2.2ohm' is ONE POINT on a boundary curve, not two independent limits. At the 10uF fitted the ESR floor is 20.1mohm; at the ~25.5uF the loop really sees it is ~7.3mohm. Paralleling MLCCs on this rail therefore INCREASES margin, against the folk rule. Residual margin is not provable from archived data – open verification item, closed by a bench test-load.

NCP1117 datasheet – C1 must sit within the shortest possible track of U2's input and ground terminals.

```
4 component(s) from module instance Power
source 1775e8f34a95-dirty
GENERATED from test-board.zen by zen-to-eeschema -- DO NOT EDIT
Small RP2350A test board (39 components, 2L, 51x21mm) used as a...
Sheet: /Power/
File: Power.kicad_sch
```

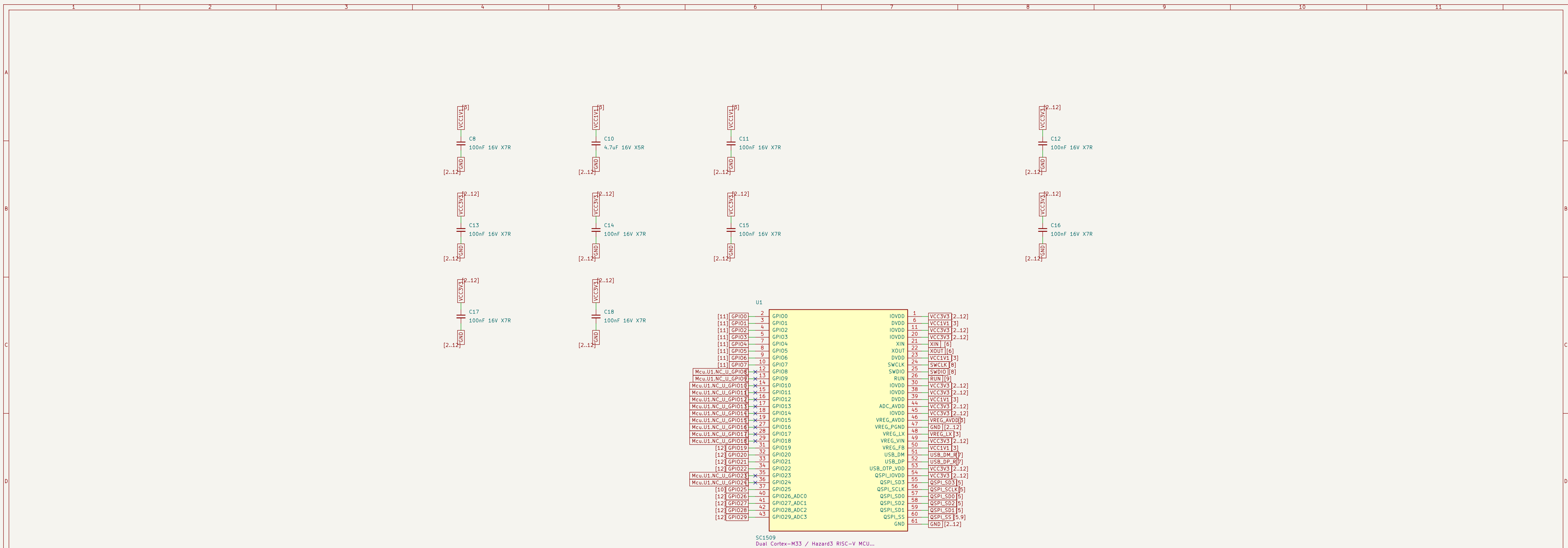
Title: test-board -- Power

Size: A3	Date: 2026-08-03T17:42:25+02:00	Rev: 1775e8f34a
KiCad E.D.A. 10.0.0		Id: 2/13



NET CLASSES ON THIS SHEET
GND -- GND_REF
GND is a B.Cu pour, not a routed trace (HW-TB-PCB24), so track width governs almost nothing. The class exists to pin GND's stitching vias to 0.6/0.3 and to give the pour an explicit identity.
VCC1V1, VCC3V3, VREG_AVDD, VREG_LX -- PWR
track width 0.4mm for loop inductance, not current capacity (HW-TB-PCB25). Broader than the design brief's section 3, which names only VCC3V3 and VBUS.

DESIGN NOTES
RP-008280-DS section 2.1: VREG_AVDD draws ~200uA through a 33 ohm filter resistor. The IR drop has to be negligible against the 3.3V rail or the analogue supply is starved.
check: `_V_DROP < Voltage("10mV")`
HW-TB-PCB11 HARD - C6/C7/C9/R5/L1 must follow the RP-008280-DS Figure 23 geometry, adjacent to the VREG pins.
HW-TB-COMP03 - L1 is polarity-marked and its orientation is NOT automated. A reversed part couples into VREG_FB.
F4 - VREG_PGND is on the GND net. RP-008280-DS 2.1 requires its switching current to return directly to PGND as a star point without disturbing the rest of GND. That is a ROUTING obligation this netlist cannot express.
F5 - hw-spec 8.4 asks for a DVDD test point; none is fitted. Probe the C7 or C10 pads, or L1 pad 2.



GND -- GND_REF
GND is a 8Cu_puff, not a routed trace (HW-TB-PCB24), so track width governs almost nothing. The class exists to pin GND's stitching via to 0.6/0.3 and to give the power an explicit identity.
VCC1, VCC2, VREG_AVDD, VREG_LV -- PWR
track width 0.4mm for loop inductance, not current capacity (HW-TB-PCB25). Broader than the design brief's section 3, which names only VCC3V3 and VBUS.
QSPI_SCLK, QSPI_S0M0, QSPI_S0M1, QSPI_S0D0, QSPI_S0D1, QSPI_S0M5 -- QSPI_BUS
track width 0.6mm, inside HW-TB-PCB18's 0.15-0.2mm band.
USB_DM_R, USB_DP_R -- USB_FS_DIFF
target 90 ohm differential
90 ohm differential target, 0.8mm / 0.5mm gap (HW-TB-PCB13 HARD). Design-intent-by-geometry: no inductance or capacitance and KIC's project schema has no impedance file -- this text is the only record. Clearance is 0.16mm, neglecting NOT HW-TB-PCB14's 2.4mm = PCB14 is a track-to-trace 3W rule and lies in layout.kicad.drc; as a net class it is unsatisfiable at the USB-C land pattern and cost 15 of 44 routed nets.
XIN, XOUT -- XTAL_12M
clearance 0.45mm is a 3x keep-out from digital and switching nets (HW-TB-PCB23 HARD), not a fab minimum -- the one class whose clearance encodes design intent rather than a manufacturing floor.

DESIGN NOTES

HW-TB-COMP06 (D15) 5 supply pins covered by 13 capacitors, 100% coverage. Pins 44/45 share an AC coupling capacitor, pins 53/54 share another, both are R_{PI}=00E0B20, 2.2Ω, own documented sharing agreement. C10 is the 4.7uF at DVDD, deviation (b), and belongs on the side of the package

POSITIVE the regulator.

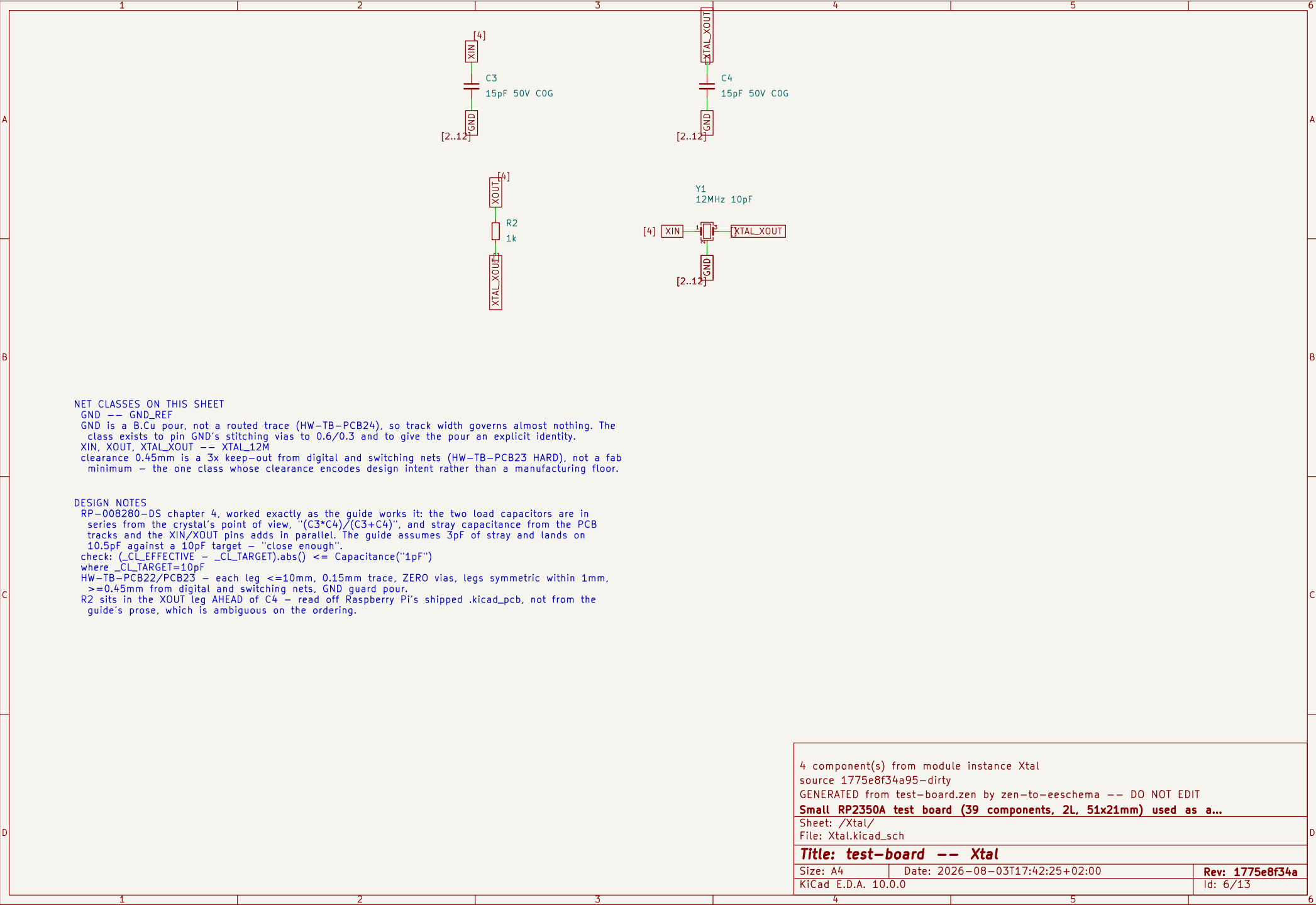
Ask the customer to provide a schematic per PIN, never per capacitor. The eight VCCV33 decouplers are one net and one MPN and are electrically interchangeable, so which pin a cap is NAMED for is a "capture-time" convention, not a fact about the board. Only coverage is the requirement.

ACCOUPLING (p44, 45) 44, 45 3.3V 2.2Ω 00E0B20 matching the R_{PI} reference, which fits no ADC/AVDD filter. Consequence: NO ADC ACCURACY CLAIM can be made for this board.

Rail ranges are declared on the I/O_{AVDD} rather than asserted in a check(); an io(Power) does not surface the caller's value at module-evaluation time, so such a check would pass unconditionally. Power = 1.8V or more than none, IOVDD 1.8-3.3V is the package range; this board runs 3.3V, DVDD is 1.1V +7-3%.

```
11 component(s) from module instance Mcu
source 1775e8f34a95-dirty
GENERATED from test-board.zen by zen-to-eeschema -- DO NOT EDIT
Small RP2350A test board (39 components, 2L, 51x21mm) used as a...
Sheet: /Mcu/
File: Mcu.kicad_sch
```

Title: test-board -- Mcu		
Size: A2	Date: 2026-08-03T17:42:25+02:00	Rev
KiCad E.D.A. 10.0.0		Id:



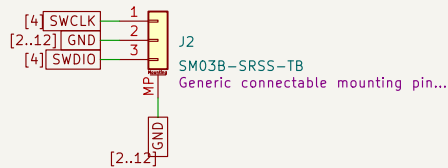
NET CLASSES ON THIS SHEET

GND -- GND_REF
GND is a B.Cu pour, not a routed trace (HW-TB-PCB24), so track width governs almost nothing. The class exists to pin GND's stitching vias to 0.6/0.3 and to give the pour an explicit identity.
XIN, XOUT, XTAL_XOUT -- XTAL_12M
clearance 0.45mm is a 3x keep-out from digital and switching nets (HW-TB-PCB23 HARD), not a fab minimum -- the one class whose clearance encodes design intent rather than a manufacturing floor.

DESIGN NOTES

RP-008280-DS chapter 4, worked exactly as the guide works it: the two load capacitors are in series from the crystal's point of view, " $(C3 \cdot C4) / (C3 + C4)$ ", and stray capacitance from the PCB tracks and the XIN/XOUT pins adds in parallel. The guide assumes 3pF of stray and lands on 10.5pF against a 10pF target -- "close enough".
check: $(_CLEFFECTIVE - _CL_TARGET).abs() \leq \text{Capacitance}("1pF")$
where $_CL_TARGET=10pF$
HW-TB-PCB22/PCB23 -- each leg $\leq 10mm$, 0.15mm trace, ZERO vias, legs symmetric within 1mm, $\geq 0.45mm$ from digital and switching nets, GND guard pour.
R2 sits in the XOUT leg AHEAD of C4 -- read off Raspberry Pi's shipped .kicad_pcb, not from the guide's prose, which is ambiguous on the ordering.

4 component(s) from module instance Xtal		
source 1775e8f34a95-dirty		
GENERATED from test-board.zen by zen-to-eeschema -- DO NOT EDIT		
Small RP2350A test board (39 components, 2L, 51x21mm) used as a...		
Sheet: /Xtal/		
File: Xtal.kicad_sch		
Title: test-board -- Xtal		
Size: A4	Date: 2026-08-03T17:42:25+02:00	Rev: 1775e8f34a
KiCad E.D.A. 10.0.0		Id: 6/13



NET CLASSES ON THIS SHEET

GND -- GND_REF

GND is a B.Cu pour, not a routed trace (HW-TB-PCB24), so track width governs almost nothing. The class exists to pin GND's stitching vias to 0.6/0.3 and to give the pour an explicit identity.

DESIGN NOTES

F6 - RP-008189-DS's 100 ohm source-termination resistors on SWCLK/SWDIO are 'highly recommended' but NOT fitted: the part budget is fixed at 39 and RPi's own RP2350 Minimal omits them too. What is given up is slew limiting and ESD/short-circuit current limiting on the two debug pins.

1 component(s) from module instance Swd

source 1775e8f34a95-dirty

GENERATED from test-board.zen by zen-to-eeschema -- DO NOT EDIT

Small RP2350A test board (39 components, 2L, 51x21mm) used as a...

Sheet: /Swd/

File: Swd.kicad_sch

Title: test-board -- Swd

Size: A4

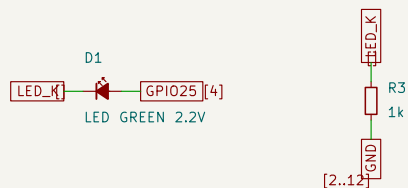
Date: 2026-08-03T17:42:25+02:00

Rev: 1775e8f34a

KiCad E.D.A. 10.0.0

Id: 8/13





NET CLASSES ON THIS SHEET

GND is a B.Cu pour, not a routed trace (HW-TB-PCB24), so track width governs almost nothing. The class exists to pin GND's stitching vias to 0.6/0.3 and to give the pour an explicit identity.

DESIGN NOTES

Two independent bounds: the APT2012SGC's 25mA absolute maximum forward current, and a floor that keeps the indicator actually visible.

_VF and _R_SERIES are passed INTO the instantiations below rather than repeated there as string literals. That is what makes these checks bind: with independent literals, editing 'value=' changed the netlist and the check went on passing over its own private constants – a consistency assertion about nothing.

V_f is the datasheet's 2.2V typ AT IF=20mA, ~18x this operating point. A GaP green die sits materially below its 20mA V_f at ~1mA, so the true current is HIGHER than the computed figure; using V_f max 2.5V instead gives 0.80mA. The honest reading is a band of roughly 0.8–1.5mA, not the point value hw-spec.md's power budget calls "derived". Both bounds below hold across it.

```
check: _I_LED < Current("25mA")
```

```
check: _I_LED > Current("0.5mA")
```

F7 - GPIO25 is Pico's own LED pin, so stock Pico firmware blinks this board unchanged. The cost is a cross-board run from QFN side C to the short edge. GPIO16/17/18 on side B are free alternates if placement gets tight.

2 component(s) from module instance Led

GENERATED from test-board.zen by zen-to-eeschema -- DO NOT EDIT

Small RP2350A test board (39 components, 2L, 51x21mm) used as a...

Sheet: /Led/

File: Led.kicad_sch

Title: test-board -- Led

Size: A4

Date: 2026-08-03T17:42:25+02:00

Rev: 1775e8f34a

KiCad E.D.A. 10.0.0

Id: 10/13

